

Copyright Filing Application Draft — UpTrend

1. Title of the Work

UpTrend – AI-Powered Stock Market Learning, Portfolio Simulation, and Paper Trading Platform

Abstract

UpTrend is a MERN-stack based web platform developed for stock market learning, portfolio simulation, and paper trading. The system integrates educational modules, AI-assisted guidance, market visualization, and virtual trading into a unified platform. Users can learn trading concepts, execute virtual buy/sell operations using simulated capital, monitor portfolio performance, and analyze market trends through interactive dashboards and charts.

Objective

The objective of UpTrend is to provide beginners with a risk-free environment for learning stock market concepts and practicing trading strategies using simulated trading tools and educational resources.

Core Modules

- User Authentication
- Dashboard Analytics
- Educational Academy
- Portfolio Management
- Paper Trading Engine
- Market Visualization
- AI Assistance

Operational Workflow

1. User logs into the platform.
2. Dashboard loads portfolio and learning data.
3. User accesses educational modules and quizzes.
4. User performs virtual buy/sell operations.
5. Portfolio metrics and market charts update dynamically.
6. AI assistant provides contextual learning guidance.
7. Trade history and analytics are stored in the database.

System Architecture

Frontend:

- React.js
- Tailwind CSS

Backend:

- Node.js
- Express.js

Database:

- MongoDB

External Services:

- Market Data APIs
- AI Integration Modules

Pseudocode:

Dashboard

BEGIN

FETCH user_data

FETCH portfolio_data

FETCH academy_progress

CALCULATE xp_percentage

CALCULATE quiz_accuracy

DISPLAY dashboard_widgets

DISPLAY portfolio_summary

END

Academy

BEGIN

AUTHENTICATE user

LOAD learning_modules

FOR each module

 CALCULATE completion_percentage

 DISPLAY progress

END FOR

UPDATE module_progress

SAVE progress

END

Portfolio

BEGIN

LOAD trade_history

CALCULATE total_profit_loss

CALCULATE win_rate

DISPLAY portfolio_metrics

DISPLAY equity_chart

END

Trading Engine

BEGIN

LOAD stock_data

UPDATE live_market_price

IF BUY order THEN

 CREATE long_position

END IF

IF SELL order THEN

 CREATE short_position

END IF

UPDATE portfolio

STORE trade_history

END

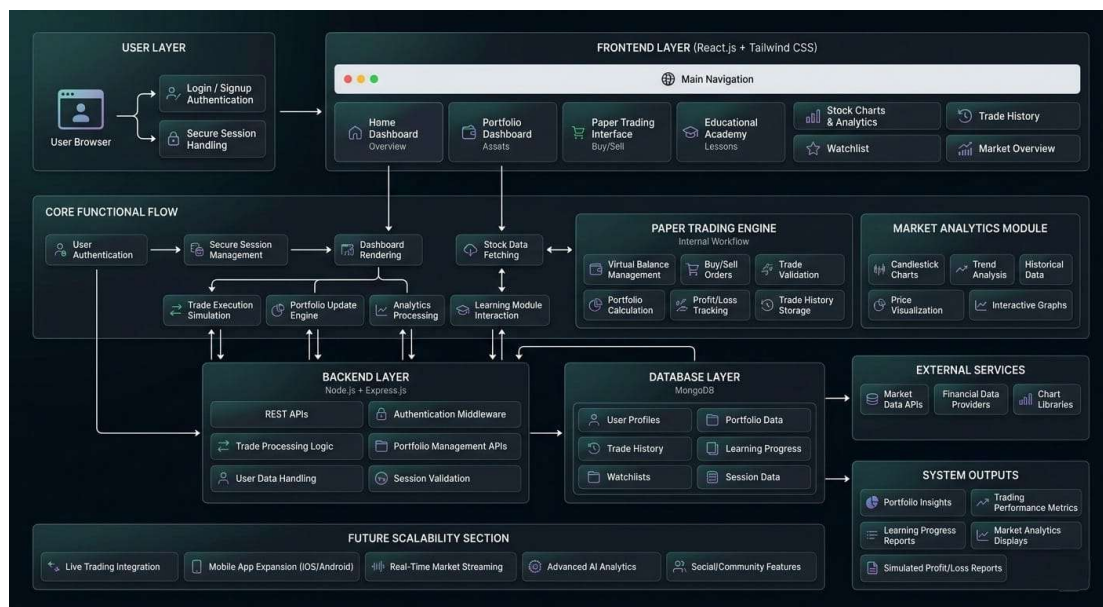


Fig 1: System Architecture of UpTrend

Future Enhancements

- Real-time market integration
- AI-based trading analytics
- Mobile application support

Language of the Work

- JavaScript
- React.js
- Node.js
- Express.js
- MongoDB
- HTML
- CSS
- Tailwind CSS

Whether the Work Has Been Published Already?

No

Inventor details

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In case of Computer Software, please submit source code/ programing of the software in pdf format as well as executable format.

Note: Please email the duly filled and signed form to ipr@ddn.upes.ac.in

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